THE MASTERMIND COACHING CLASSES

Revision Notes

Class 9 · Maths

Ganita Manjari — Part 2, Chapters 9 to 14 · CBSE / RBSE Syllabus

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
9	Propositions and their Converses	Propositions, converses, counterexamples, if-and-only-if ideas
10	How Quantities Combine: Understanding Data	Weighted average, mixtures, stacked and 100% stacked bar charts
11	The World of Algorithms	Algorithms, divisors, gcd, Euclid's subtraction and Āryabhaṭa's division algorithm
12	Quadrilaterals	Quadrilaterals, parallelogram tests, Midpoint Theorem, Centroid Theorem, Varignon parallelogram, tiling
13	Two Variables, One Line	Linear equations in two variables, slope, pair of equations, substitution, elimination and graphical methods
14	Math of Space: Surface Area and Volume	Surface area and volume of cuboid, cube, cylinder, cone, pyramid, sphere and hemisphere; estimation

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CHAPTER 09 · MATHS (PART 2)

Propositions and their Converses

Mathematics is built on statements that can be proved or disproved. This chapter shows how to state a claim as an 'if … then …' proposition, how to write its converse, why a true proposition may have a false converse, and how a single counterexample can bring down a claim.

Big Questions

- What is a proposition and what is its converse?
- If a proposition is true, must its converse be true?
- How do we disprove a claim, and how do we prove that a converse holds?

9.1 Propositions and Converses

DEFINITION

A **proposition** is a statement that is either **true or false**. Most mathematical propositions have the form **'If X then Y'**, also written **'X implies Y'**.

The **converse** of 'If X then Y' is '**If Y then X**': the 'if' part and the 'then' part are swapped. The same idea can be worded as 'Y when X', 'X only if Y' or 'X implies Y'; these all say the same thing.

Proposition	Converse	True / False
If two sides of a triangle are equal, the angles opposite them are equal	If two angles of a triangle are equal, the sides opposite them are equal	Both true
If it rains, the road is wet	If the road is wet, it has rained	Proposition true; converse false (a tanker may spill water)
If a number is a multiple of 6, it is a multiple of 3	If a number is a multiple of 3, it is a multiple of 6	Proposition true; converse false (9)
If two triangles have the same area, they are congruent	If two triangles are congruent, they have the same area	Proposition false; converse true

IMPORTANT

A true proposition may have a **true or a false** converse. It can even happen that **both are false**. The truth of a converse must always be checked separately.

9.2 Counterexamples

DEFINITION

A **counterexample** is a single instance in which the 'if' part of a proposition is **true** but the 'then' part is **false**. One counterexample is enough to show that a proposition is **false**.

- 'If a number is odd, then it is prime': **9** is odd but not prime.
- 'All numbers of the form $n^2 + n + 11$ are prime': for n = 11 the value is 143 = 11 × 13.

- ‘If n is prime, then $2^n - 1$ is prime’: for n = 11 the value is 2047 = 23 × 89.
- Fermat claimed that every number $2^{2^n} + 1$ is prime (3, 5, 17, 257, …). Euler showed that for n = 5 the number $2^{32} + 1 = 641 \times 6700417$ is composite.

> **REMEMBER**
> Checking many cases **supports** a claim but never **proves** it. A proof must cover every case; a disproof needs just one counterexample.

9.3 Number-Theory Examples

- **Multiples:** a multiple of 6 is a multiple of 3 (6 = 3 × 2), but not conversely.
- **Divisibility by a product:** if n is divisible by 24, it is divisible by 4 and by 6. The converse is **false**: 12 is divisible by 4 and 6 but not by 24, because 4 and 6 share a factor.
- **Divisible by 60 ⇔ divisible by 5 and 12:** here the converse is also true since 5 and 12 have no common factor and 5 × 12 = 60.
- **Digit sum rule:** a number is divisible by 3 exactly when the digit sum is a multiple of 3 (both directions true).
- **Divisible by 8:** being divisible by 2 and 4 is **not enough** (12 is not divisible by 8).

▸ Factors and perfect squares

Every factor has a **partner** whose product with it is the number. For 12 the pairs are (1, 12), (2, 6), (3, 4): six factors. A pair that repeats a number, such as (6, 6) for 36, contributes only **one** factor.

Statement	Direction	True?
Perfect square ⇒ odd number of factors	Proposition P	True
Odd number of factors ⇒ perfect square	Converse Q	True
Square of a prime p^2 has exactly 3 factors (1, p, p^2)	Proposition	True; converse also true
Product of two unequal primes pq has exactly 4 factors (1, p, q, pq)	Proposition	True; converse false (8 has 4 factors)

> **NOTE**
> A number has **at most one** repeated factor pair (f, f). So a square number has exactly one unpaired factor, which makes the count **odd**.

9.4 Geometry Examples

- **Parallel lines:** if two lines are parallel, corresponding angles are equal; the converse is true and is used to prove lines parallel.
- **Square ⇒ all angles equal:** converse false (a rectangle).
- **Rectangle ⇒ equal diagonals:** converse false (an isosceles trapezium has equal diagonals).
- **Equal sides ⇒ equal angles:** converse true (draw the altitude and use congruence).

▸ Baudhāyana–Pythagoras theorem and its converse

FORMULA

Theorem: if a triangle with sides a, b, c is right-angled (hypotenuse c), then $a^2 + b^2 = c^2$. **Converse:** if $a^2 + b^2 = c^2$, then the triangle is right-angled.

Step 1 — Set up

Let triangle ABC have sides a, b, c with $a^2 + b^2 = c^2$.

Step 2 — Build a right triangle

Construct triangle XYZ with perpendicular sides YZ = a and XZ = b.

Step 3 — Find XY

By the theorem, $XY^2 = a^2 + b^2 = c^2$, so XY = c.

Step 4 — Compare

Both triangles have sides a, b, c, so they are congruent by SSS.

Step 5 — Conclude

Angle C equals angle Z = 90°, so triangle ABC is right-angled: the converse is true.

Sides	$a^2 + b^2$	c^2	Right-angled?
3, 4, 5	25	25	Yes
8, 15, 17	289	289	Yes
6, 8, 11	100	121	No
9, 12, 16	225	256	No

Quick Revision: Likely Exam Questions

Likely question	Model answer
Define a proposition and its converse.	A proposition is a statement that is true or false. The converse of 'If X then Y' is 'If Y then X'.
If a proposition is true, is its converse always true?	No. 'If a number is a multiple of 6, it is a multiple of 3' is true but its converse is false (9).
What is a counterexample?	An instance where the 'if' part holds but the 'then' part fails; one such case disproves a proposition.
Give a counterexample to '$n^2 + n + 11$ is always prime'.	n = 11 gives 143 = 11 × 13.
State the converse of the Baudhāyana–Pythagoras theorem.	If $a^2 + b^2 = c^2$ in a triangle with sides a, b, c, then the triangle is right-angled.
Which proof technique shows the converse of Pythagoras is true?	Build a right triangle with legs a and b; its hypotenuse is c; SSS makes the triangles congruent.
How many factors does a perfect square have?	An odd number, because one factor pair repeats (f, f).
Is 'divisible by 2 and 4' enough for divisibility by 8?	No; 12 is divisible by 2 and 4 but not by 8.

Likely question	Model answer
Can both a proposition and its converse be false?	Yes, e.g. ‘If a number is a multiple of 3, it is a multiple of 5’ and its converse.
Write ‘A number has an odd number of factors when it is a perfect square’ in if–then form.	If a number is a perfect square, then it has an odd number of factors.

★ KEY TAKEAWAYS

- A **proposition** is true or false; its **converse** swaps the ‘if’ and ‘then’ parts.
- A true proposition can have a **true or a false converse**; **check each separately**.
- **One counterexample** disproves a proposition; many examples never prove one.
- **Perfect square ⇔ odd number of factors**; p^2 has 3 factors; pq has 4.
- The **converse of the Baudhāyana–Pythagoras theorem** is true and is proved by SSS congruence.

CHAPTER 10 · MATHS (PART 2)

How Quantities Combine: Understanding Data

Averages of different groups cannot always be averaged again. This chapter shows how to combine data of different sizes or importance using the weighted mean, how it applies to mixtures and marks, and how stacked and 100% stacked bar charts display totals and shares.

Big Questions

- How do we combine the averages of groups of different sizes?
- How does the weighted mean apply to mixtures, marks and ratings?
- What can and cannot be read from stacked and 100% stacked bar charts?

10.1 Average of Averages

If group 1 has **n** values with average **a** and group 2 has **m** values with average **b**, then the total of group 1 is **an** and of group 2 is **bm**.

FORMULA

Combined average = (a × n + b × m) ÷ (n + m). For three groups: (ap + bq + cr) ÷ (p + q + r).

Case	Method	Why
Groups of equal size	Add the averages and divide by the number of groups	Every group has the same weight
Groups of different size	Use the weighted formula	A larger group must count for more

Step 1 — Find each total
Multiply each average by the size of its group.

Step 2 — Add the totals
This gives the sum of all the values.

Step 3 — Divide by total count
Divide by n + m to get the combined average.

REMEMBER

Example: 24 students average 68 and 16 students average 78. Combined = (24 × 68 + 16 × 78) ÷ 40 = 72. The simple average (73) is wrong because the groups are unequal.

10.2 Mixtures

Concentration = quantity of the substance ÷ quantity of the mixture. When mixtures are combined, the concentration is the **weighted average** of the concentrations, the weights being the **quantities**.

FORMULA

Concentration $x = (w_1x_1 + w_2x_2 + \ldots + w_nx_n) \div (w_1 + w_2 + \ldots + w_n)$ — the **weighted mean**.

- **Equal quantities:** the concentration lies **midway**, e.g. 10% and 20% jaggery give 15%.
- **Unequal quantities:** the result is **closer to the concentration of the larger quantity**.
- The mixture's concentration always lies **between** the smallest and largest concentrations mixed.
- To find an unknown quantity y, write the weighted-mean equation and solve it.

▸ Worked example — missing quantity

How much 25% solution must be added to 200 mL of 10% solution to get 15%? (20 + 0.25y) ÷ (200 + y) = 0.15 ⇒ 20 + 0.25y = 30 + 0.15y ⇒ **y = 100 mL**.

▸ Dilution

Adding plain water does not change the amount of solute. If 12 litres of 10% solution is diluted to 8%: solute = 1.2 litres; new volume = 1.2 ÷ 0.08 = 15 litres; water added = **3 litres**.

> **NOTE**
> Brahmagupta (c. 628 CE) used this formula to find the **mean depth of an irregular pit**; Śrīdharācārya (c. 750 CE) used it for the **purity of gold alloys** (varṇa; 16 varṇa = pure gold = 24 karat).

10.3 Custom Weights

Weights need not be physical weights. They show **how important** each part is: marks in internals, project and final exam in the ratio 3 : 2 : 5, or ratings of food, service and ambience.

Component	Score	Weight	Score × weight
Internals	58	3	174
Project	66	2	132
Final exam	75	5	375
Total		10	681

Weighted score = 681 ÷ 10 = **68.1%**. The simple average (66.33%) is lower because the best score carries the highest weight.

- Convert each score to the same scale (percentage) **before** weighting.
- A 'recent reviews count double' rule is also a weighted mean.
- Adding one new value x to n values with average a: new average = (n × a + x) ÷ (n + 1).

10.4 Visualising Data: Stacked Bar Charts

A **stacked bar chart** places the parts of a total end to end in one bar. The full bar shows the **total**; each segment shows a component.

Chart	Good for	Limitation
Cluster-column	Comparing categories across groups or groups across categories	Totals are hard to see
Stacked bar	Comparing **totals** while keeping the parts	Segments after the first do not start at 0, so they are harder to compare
100% stacked bar	Comparing **shares** (percentages) of parts	Does **not** show the actual amounts

10.5 100% Stacked Bars

In a 100% stacked bar every bar has the **same length** (100%). Each segment is the part's percentage of that bar's own total. It is similar to a pie chart, but lengths are easier to compare than angles.

Step 1 — Total
Add the components of each bar.

Step 2 — Share
Divide each component by the total and multiply by 100.

Step 3 — Draw
Make each bar equal in length and divide it in proportion to the shares.

IMPORTANT

From a 100% stacked bar you **can** compare parts **within** one bar. You **cannot** compare actual amounts **across** bars, since the totals are unknown. Example: Bakery A bread 45% and Bakery B bread 30% does not mean A sold more bread; if B's total is three times A's, B sold more.

NOTE

Averages of a diverse population can hide differences. Breaking data into age groups, gender or region (as in time-use charts) gives a clearer picture.

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the formula for the combined average of two groups.	(a × n + b × m) ÷ (n + m), where a, b are averages and n, m the sizes.
When can we simply average the group averages?	When all the groups have the same size.
What is the weighted mean?	(w1x1 + w2x2 + …) ÷ (w1 + w2 + …), an average in which each value counts by its weight.
Mix 400 mL of 5% and 100 mL of 10% solution. Find the concentration.	(400 × 5 + 100 × 10) ÷ 500 = 6%.
Average of 15 values is 62. A new value 78 is added. New average?	(15 × 62 + 78) ÷ 16 = 63.
What does the length of a full stacked bar show?	The total of all the components.
How is a 100% stacked bar different from a stacked bar?	Every bar has the same length and shows only shares, not actual values.
Why can't we compare amounts across bars in a 100% stacked chart?	The totals of the bars may differ and are not shown.
Who used the weighted-mean formula for the depth of a pit?	Brahmagupta, in the Brāhmasphuṭasiddhānta (c. 628 CE).
Why does the final exam raise a student's score more under weights 3 : 2 : 5 than under equal weights?	It has the largest weight, so its score influences the average most.

★ KEY TAKEAWAYS

- Combined average = **(an + bm) ÷ (n + m)**; equal sizes ⇒ simple average of the averages.
- The **weighted mean** (Σ weight × value ÷ Σ weights) applies to mixtures, marks, ratings and pit depths.
- In mixtures the **quantities are the weights**; the result lies between the smallest and largest concentration.
- A **stacked bar** shows totals and parts; a **100% stacked bar** shows shares only.
- From a 100% stacked bar you **cannot** infer actual amounts across bars.

CHAPTER 11 · MATHS (PART 2)

The World of Algorithms

An algorithm is a precise, step-by-step recipe for solving a problem. This chapter builds algorithms for adding numbers and for finding the gcd, shows how each version is made faster, and ends with the efficient division method known to Āryabhaṭa.

Big Questions

- What is an algorithm and how do we describe one?
- How can we justify that an algorithm is correct?
- How do we judge and improve its efficiency?

What Is an Algorithm?

DEFINITION

An **algorithm** is a systematic, step-by-step procedure to solve a problem. It is built from **basic steps** we can already do directly (for example, adding two single digits or checking if a smaller number divides a larger one).

- Steps may be **repeated**: *for each j from 1 to n, check whether j divides n*.
- Steps may be **conditional**: *if j divides n, add j to the list*.
- We give names (like **j**, **carry**, **list-of-divisors**) to the values we keep track of.
- **Executing** an algorithm means running through its steps one by one.
- We must always **justify** that an algorithm is correct and can **analyse** how many steps it takes.

Adding Numbers Digit by Digit

Step 1 — Align
Write the numbers one below the other, digits aligned from the right.

Step 2 — Units column
Add the rightmost digits; if the sum is less than 10 write it and set carry = 0, otherwise write the units digit and set carry = 1.

Step 3 — Next column
Add the two digits and the current carry; write the units digit; set the new carry.

Step 4 — Repeat
Repeat Step 3 until no digits are left.

Step 5 — Final carry
If carry = 1, write 1 to the left of the answer.

- The carry is never more than **1**, because the largest column total is 9 + 9 + 1 = 19.
- Without Step 5, 95 + 17 would be written as 12 instead of **112**.
- **Why it works:** each number is split into units, tens, hundreds…; each group is added separately and a group of ten is carried to the next place.

Method	3-digit → 5-digit numbers	Effect
Counting dots	100 times as many dots	Work grows with the **value**
Column algorithm	Only 2 more columns	Work grows with the **number of digits**

> **NOTE**
> With 1 second per dot, counting 3347 + 2729 takes over 1 hour 40 minutes; the column method with 15 seconds per column takes about 1 minute.

Divisors of a Number

Step 1
Start with an empty list-of-divisors.

Step 2
For each j in 1, 2, 3, …, n: if j divides n, add j at the end of the list.

- divisors(18) = **[1, 2, 3, 6, 9, 18]**; the list is automatically in **increasing order**.
- divisors(375) = [1, 3, 5, 15, 25, 75, 125, 375]; divisors(825) = [1, 3, 5, 11, 15, 25, 33, 55, 75, 165, 275, 825].
- A **list** is a simple **data structure**: it organises values so the algorithm can use them later.

Greatest Common Divisor — First Algorithm

Step 1–2
Compute divisors-of-m and divisors-of-n.

Step 3
Start with an empty list common-divisors.

Step 4
For each x in divisors-of-m, if x is also in divisors-of-n, add x to common-divisors.

Step 5
Report the rightmost element as gcd(m, n).

For 375 and 825 the common divisors are [1, 3, 5, 15, 25, 75], so **gcd = 75**. The rightmost element is the largest because the lists are in increasing order.

Three Improvements

Improvement	What changes	Numbers checked
Combine the two scans	Run j from 1 to max(m, n) once, testing both numbers	max(m, n) instead of m + n
Directly find common divisors	Check j up to min(m, n), as a common divisor cannot exceed the smaller number	min(m, n)
Drop the list	Keep only **most-recent-common-divisor** (start at 1); for k = 2 … min(m, n), if k divides both, update it to k	min(m, n) − 1, no list needed

IMPORTANT

Even the last version scans from 1 to min(m, n). If the smaller number goes from 3 to 4 digits, the work becomes **10 times** as large; from 3 to 5 digits, **100 times**. The effort depends on the value, not the number of digits.

Euclid's Subtraction Algorithm

FORMULA

For m ≥ n: **gcd(m, n) = gcd(n, m − n)**, and **gcd(m, 0) = m**.

▸ Why gcd(m, n) = gcd(n, m − n)

- If d divides m and n, write m = ad, n = bd; then m − n = (a − b)d, so d divides m − n.
- If d divides n and m − n, write n = xd, m − n = yd; then m = n + (m − n) = (x + y)d, so d divides m.
- So both pairs have the **same common divisors** and therefore the same gcd.

Step 1

If m < n, reverse the numbers and compute gcd(n, m).

Step 2

If n = 0, report the answer m.

Step 3

Otherwise reduce the problem to gcd(n, m − n).

gcd(375, 825): (825, 375) → (375, 450) → (450, 375) → (375, 75) → (75, 300) → … → (75, 75) → (75, 0). Step 3 is used **7 times** and the gcd is **75**.

NOTE

gcd(99, 2) needs about 50 reductions (99 → 97 → 95 …) because we subtract 2 again and again. In general gcd(2k + 1, 2) needs about k reductions: effort proportional to the **value**.

Āryabhaṭa's Division Algorithm

Repeated subtraction of n from m ends with the remainder m mod n. Replace it with one division (the ‘long division’ method of Āryabhaṭa, 499 CE):

FORMULA

gcd(m, n) = gcd(n, m mod n); stop when n = 0 (the answer is m). Example: 17 mod 5 = 2, 33 mod 7 = 5.

Example	Chain of reductions	Answer
gcd(99, 2)	(2, 1) → (1, 0)	1 (2 reductions)
gcd(825, 375)	825 = 375×2 + 75; 375 = 75×5 + 0	75
gcd(60, 16)	60 = 16×3 + 12; 16 = 12×1 + 4; 12 = 4×3 + 0	4

- The number of reductions now depends on the **number of digits**, not on the value (proved in later grades).
- Al-Khwārizmī (c. 820 CE) wrote on Indian numerals; his name in Latin, *Algoritmi*, gave the word **algorithm**.

Primes, Divisors and Factorisation

- **prime(n):** compute divisors(n); n is prime if the list has exactly two entries (1 and n).
- **primedivisors(n):** compute divisors(n) and keep only those that are prime.
- Prime factorisation $180 = 2^2 \times 3^2 \times 5^1$ can be stored as the list of pairs [(2, 2), (3, 2), (5, 1)]; gcd = product of each common prime to the **smaller** power.

Likely question	Model answer
What is an algorithm?	A precise step-by-step procedure to solve a problem, made of basic steps, repeats and conditions.
Why can the carry never exceed 1?	The largest column total is 9 + 9 + 1 = 19, which is 9 plus one ten.
Why do we check common divisors only up to min(m, n)?	A common divisor cannot be larger than the smaller number.
Why is Euclid's division form faster than the subtraction form?	One division removes all the repeated subtractions at once, so the steps depend on the digits, not the value.

★ KEY TAKEAWAYS

- An **algorithm** is a correct, step-by-step procedure built from basic steps.
- Steps can be repeated (**for each**) or conditional (**if**); intermediate values are named.
- Always **justify** correctness and **analyse** the number of steps.
- Column addition: work grows with the number of digits; counting grows with the value.
- gcd by lists: divisors → common divisors → rightmost element.
- Euclid: **gcd(m, n) = gcd(n, m − n)**; Āryabhaṭa: **gcd(m, n) = gcd(n, m mod n)**, stop at n = 0.
- The division form needs far fewer reductions (e.g. gcd(99, 2): 2 instead of about 50).

CHAPTER 12 · MATHS (PART 2)

Quadrilaterals

Quadrilaterals appear in floors, frames and machines. This chapter defines them precisely, proves the tests that tell us a quadrilateral is a parallelogram, uses parallelograms to prove the Midpoint and Centroid Theorems, and shows that every quadrilateral can tile the plane.

Big Questions

- What exactly is a quadrilateral?
- How can we recognise a parallelogram?
- How do parallelograms help prove facts about triangles?
- Which quadrilaterals can tile the plane?

What Is a Quadrilateral?

DEFINITION

Four distinct points A, B, C, D in a plane, with segments AB, BC, CD, DA, form **quadrilateral ABCD** if every point of these segments other than A, B, C, D lies on **exactly one** of them. So no three vertices are collinear, the figure is planar, and the sides do not cross.

- **Vertices:** A, B, C, D. **Sides:** AB, BC, CD, DA. **Diagonals:** AC and BD.
- **Adjacent** sides share a vertex; **opposite** sides do not (AB and CD).
- ABCD can also be named BCDA, CDAB, DABC, DCBA, ADCB, BADC, CBAD — but not ACBD.
- Angle sum of a quadrilateral = **360°** (split by a diagonal into two triangles), even if it is non-convex.

Type	Test	Diagonals
Convex	All internal angles < 180°	Intersect
Non-convex ('dart')	One internal angle > 180°	Do not intersect
Self-intersecting	Two sides cross (not a quadrilateral by definition; occurs in 4-bar linkages)	—

Properties of a Parallelogram

DEFINITION

A **parallelogram** is a quadrilateral whose opposite sides are parallel. Rectangles, rhombuses and squares are parallelograms.

Theorem 1 — property	Proof idea
(a) Opposite sides are equal	Diagonal AC makes △ACD ≅ △CAB by ASA
(b) Opposite angles are equal	Adjacent angles add to 180° (co-interior), so ∠A = ∠C and ∠B = ∠D
(c) Diagonals bisect each other	△AED ≅ △CEB by ASA/AAS

Tests for a Parallelogram (Converses)

Theorem	Statement	Proof idea
2	Opposite sides equal ⇒ parallelogram	△ACD ≅ △CAB (SSS), so alternate angles equal
3	Opposite angles equal ⇒ parallelogram	2(∠A + ∠B) = 360°, so ∠A + ∠B = 180° ⇒ sides parallel
4	Diagonals bisect each other ⇒ parallelogram	△AED ≅ △CEB (SAS) ⇒ AD ∥ BC; △EAB ≅ △ECD ⇒ AB ∥ DC
5	One pair of opposite sides **equal and parallel** ⇒ parallelogram	△EAB ≅ △ECD (ASA) ⇒ diagonals bisect each other

IMPORTANT

One pair of equal opposite sides is not enough — an isosceles trapezium has one equal pair but is not a parallelogram. The equal sides must also be parallel.

- A parallelogram with one right angle is a **rectangle**; with equal diagonals also a rectangle.
- If the diagonal AC of a parallelogram bisects ∠A, it also bisects ∠C and ABCD is a **rhombus**.
- A rhombus with a right angle (or equal diagonals) is a **square**; all rhombus diagonals are perpendicular.

Midpoint Theorem and Its Converse

FORMULA

Midpoint Theorem (Theorem 6): in △ABC, if P and Q are the midpoints of AB and AC, then **PQ ∥ BC** and **PQ = ½ BC**.

Construction
Draw the line through C parallel to BA, meeting PQ produced at R.

Congruence
△APQ ≅ △CRQ (AAS), so CR = AP = BP and PQ = QR.

Parallelogram
BCRP has CR ∥ BP and CR = BP, so it is a parallelogram (Theorem 5).

Conclusion
PQ ∥ BC and PQ = ½ PR = ½ BC.

FORMULA

Converse (Theorem 7): the line through the midpoint of one side parallel to another side **bisects the third side**. Proof: the midpoint M of AC gives PM ∥ BC; only one line through P is parallel to BC, so PM and PQ coincide and Q = M.

- The midpoints of the sides of a triangle make **four congruent triangles**, each similar to the original.
- Perimeter of the triangle joining the midpoints = ½ perimeter of the original.

Centroid Theorem

FORMULA

Theorem 8: the three medians of a triangle meet at a point, the **centroid G**, which divides each median in the ratio **2 : 1** with the longer part towards the vertex. So AG = ⅔ AD and GD = ⅓ AD.

Proof outline: for medians CP and BQ meeting at M, take midpoints X, Y of BM and CM. By the Midpoint Theorem PQ ∥ BC ∥ XY and PQ = XY = ½ BC, so △MPQ ≅ △MYX and CM : MP = 2 : 1 = BM : MQ. The same argument with the third median gives the same point.

Varignon Parallelogram

FORMULA

Theorem 9: the midpoints P, Q, R, S of the sides of any quadrilateral ABCD form a **parallelogram** PQRS. Reason: PQ ∥ AC ∥ SR and QR ∥ BD ∥ PS (Midpoint Theorem in triangles ABC, ADC, BCD, BAD).

If diagonals AC and BD are…	Varignon parallelogram PQRS is a…
Any	Parallelogram (sides AC/2 and BD/2)
Equal	Rhombus
Perpendicular	Rectangle
Equal and perpendicular	Square

- Perimeter of PQRS = AC + BD.
- PR and QS (diagonals of PQRS) **bisect each other**.
- If AB ∥ DC and AB ≠ DC, the midpoints G, H of AC, BD satisfy GH ∥ AB and GH = ½ |DC − AB|.

Tiling the Plane

- A **tiling** covers the plane with copies of a shape without gaps or overlaps; the angles at any vertex must add to **360°**.
- Parallelograms tile (the ‘sticks’ grid); so does **any triangle** (two copies, one turned 180°, make a parallelogram).
- **Any quadrilateral** (even non-convex) tiles: rotate a copy by 180° about the midpoint of each edge, or build on the grid of Varignon parallelograms.
- A regular pentagon cannot tile: each angle is 108° and 360 is not a multiple of 108.
- The 2023 ‘hat’ shape tiles the plane only aperiodically.

Likely question	Model answer
Name the three conditions needed for a quadrilateral.	No three vertices collinear, all in one plane, and sides that do not cross each other.
State the equal-parallel-sides test.	A quadrilateral with one pair of opposite sides both equal and parallel is a parallelogram.
In △ABC, P, Q are midpoints of AB, AC and BC = 20 cm. Find PQ.	PQ = ½ × 20 = 10 cm and PQ ∥ BC.

Likely question	Model answer
A median is 27 cm long. How far is the centroid from the vertex?	⅔ × 27 = 18 cm (and 9 cm from the midpoint of the side).
Diagonals of a quadrilateral are 8 cm and 6 cm. Perimeter of its Varignon parallelogram?	4 + 3 + 4 + 3 = 14 cm, which equals 8 + 6.

★ KEY TAKEAWAYS

- A quadrilateral has no three collinear vertices, lies in one plane and does not cross itself; its angles add to **360°**.
- Parallelogram tests: opposite sides equal, opposite angles equal, diagonals bisect each other, or **one pair of opposite sides equal and parallel**.
- **Midpoint Theorem:** PQ ∥ BC and PQ = ½ BC; its converse: a line through a midpoint parallel to a side bisects the other side.
- **Centroid Theorem:** medians are concurrent and divide in the ratio **2 : 1**.
- **Varignon:** side midpoints of any quadrilateral form a parallelogram (perimeter = sum of the diagonals).
- Angles around a vertex of a tiling add to 360°; every triangle and every quadrilateral tiles the plane.

CHAPTER 13 · MATHS (PART 2)

Two Variables, One Line

A single equation with two unknowns has endless solutions that lie on a straight line. Two such equations together can meet at one point, never meet, or lie on top of each other. This chapter shows how to read a line's slope, how to solve a pair by substitution, elimination and graphs, and how to predict the answer from the coefficients alone.

Big Questions

- What does the solution of a linear equation in two variables look like?
- What is the slope of a line and what does it tell us?
- How do we solve a pair of linear equations and how many solutions can it have?

Linear Equations in Two Variables

DEFINITION

An equation that can be written as **ax + by + c = 0**, where a, b, c are real numbers and **a and b are not both zero**, is a **linear equation in two variables**. This is its **standard form**.

- Example: mangoes ₹60/kg and bananas ₹50/kg with total ₹280 gives **60x + 50y = 280**.
- Coefficients can be fractions, but we usually multiply through to get **integer coefficients** (¼x + ⅗y − 2 = 0 becomes 5x + 12y − 40 = 0 after multiplying by 20).
- x + 7 = 0 is also linear in two variables: write it as **1·x + 0·y + 7 = 0**.

Equation	Standard form	a	b	c
3y = 5x − 7	5x − 3y − 7 = 0	5	−3	−7
x = 8	x + 0y − 8 = 0	1	0	−8
3y = 1	0x + 3y − 1 = 0	0	3	−1

Solutions and Their Graph

A **solution** is an **ordered pair (x, y)** that satisfies the equation; the order matters because the first number is the x-value. For 3x + 2y = 12, (2, 3) and (4, 0) are solutions, but (1, 3) is not because 3 + 6 = 9 ≠ 12.

- Choose any value of x, substitute and solve for y: there are **infinitely many solutions**.
- Plotting the solutions gives a **straight line**; every point of the line is a solution and every solution is a point of the line.
- Quick way to draw it: put **x = 0** to get the y-intercept point, and **y = 0** to get the x-intercept point, then join them.

Equation	x = 0	y = 0
2x + y = 7	(0, 7)	(3.5, 0)
4x + 3y = 12	(0, 4)	(3, 0)

Equation	x = 0	y = 0
2x + 5y = 0	(0, 0)	(0, 0), so take x = 5 → (5, −2)

IMPORTANT

If **c = 0**, then x = 0, y = 0 satisfies ax + by = 0, so the line **passes through the origin**. Multiplying an equation by a non-zero number does not change its solutions: 3x + 4y = 7 and 6x + 8y = 14 have the same solution set.

▸ Finding unknown constants

If (−3, 4) is a solution of 3ax + 4y = −2, substitute: −9a + 16 = −2, so **a = 2**. Substituting a known solution turns the equation into an equation in the unknown constant.

Slope of a Line

The **slope** (gradient) measures how steep a line is. Take two points A(x_1, y_1) and B(x_2, y_2) on the line; the horizontal change is the **run** and the vertical change is the **rise**.

FORMULA

Slope m = rise ÷ run = ($y_2 - y_1$) ÷ ($x_2 - x_1$). It is the same for any two points of the line and does not depend on the order in which the points are taken.

- Line through (2, 0) and (4, 3): slope = 3/2.
- Line through (−4, 5) and (−1, 2): slope = (2 − 5)/(−1 + 4) = **−1**.
- A ramp rising 2.5 m over a run of 4 m has slope 2.5/4 = 5/8.

Slope	How the line looks	Example
Positive (m > 0)	Rises left to right	y = 2x + 1
Negative (m < 0)	Falls left to right	y = −3x + 4
Zero	Horizontal, parallel to x-axis	y = 5
Undefined (run = 0)	Vertical, parallel to y-axis	x = 2

Slope-Intercept Form

FORMULA

y = mx + d where **m** is the slope and **d** is the y-intercept (the line crosses the y-axis at (0, d)).

- Why m is the slope: for points on y = mx + d, (mx_2 + d) − (mx_1 + d) = m(x_2 − x_1), so rise ÷ run = m.
- From ax + by + c = 0 with b ≠ 0: **y = −(a/b)x − c/b**, so **m = −a/b** and **d = −c/b**.
- Example: 5x + y = 3 → y = −5x + 3: slope −5 (y falls 5 for each 1 step in x), intercept 3.
- Two lines with the **same slope** and different intercepts are **parallel**.

Pair of Linear Equations

Two linear equations in the same two variables form a **pair**: $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$. A **solution of the pair** is an ordered pair that satisfies **both** equations, i.e. a point lying on both lines.

- Framing: 2 adult + 3 child tickets cost ₹600 and 3 adult + 2 child cost ₹700 gives **2x + 3y = 600** and **3x + 2y = 700**.
- Taxi fare = fixed ₹x + ₹y per km: 6 km for ₹122 and 8 km for ₹160 give x + 6y = 122, x + 8y = 160.

Solving by Substitution

Step 1
Express one variable in terms of the other from one equation (choose the easier one).

Step 2
Substitute this expression into the other equation to get an equation in one variable and solve it.

Step 3
Put the value back to find the other variable.

Step 4
Verify the pair in both original equations.

Example: 2x + y = 11 and x − 3y = −5. From the first, y = 11 − 2x. Then x − 3(11 − 2x) = −5 gives 7x = 28, so **x = 4**, **y = 3**.

Solving by Elimination

Step 1
Multiply the equations by suitable numbers so that the coefficients of one variable become equal.

Step 2
Add or subtract the equations to eliminate that variable.

Step 3
Solve the resulting equation and substitute back.

Step 4
Check in both equations.

Example: 3x + 4y = 18 and 5x − 2y = 4. Multiply the second by 2: 10x − 4y = 8. Adding: 13x = 26, so **x = 2**, then **y = 3**.

> **NOTE**
> If the coefficients of x and y are **interchanged** in the two equations (9x + 7y = 107 and 7x + 9y = 101), add them and subtract them to get x + y and x − y directly.

Modelling Word Problems

Type	Variables	Equations
Digits	tens x, units y	x + y = sum; (10x + y) ∓ (10y + x) = difference
Ages	present ages	‘n years ago/after’ shifts both ages by n
Fixed + per unit	fixed x, rate y	x + (units)·y = total (twice)
Ratios of income/expense	incomes 9x, 7x; expenses 4y, 3y	income − expense = saving for each
Geometry	length l, width w	l + w = half-perimeter; l − w = difference

IMPORTANT

Always **define the variables**, write two equations, solve, and finish by **checking against the story** (not only the equations).

How Many Solutions? The Ratio Test

Ratios	Lines	Number of solutions	Elimination result
$a_1/a_2 \neq b_1/b_2$	Intersect at one point	Unique solution	A variable remains
$a_1/a_2 = b_1/b_2 = c_1/c_2$	Coincident	Infinitely many	0 = 0
$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	Parallel	No solution	A false statement like 0 = 5

- x − y = 10 and 10x − 10y = 100: second equation is 10 × first, so infinitely many solutions.
- x − y = 10 and 10x − 10y = 101: after multiplying the first by 10 the left sides agree but 100 ≠ 101, so **no solution**.
- A pair never has exactly two solutions: it has 0, 1 or infinitely many.

Graphical Method

Step 1

Make a table of at least two points for each equation (x = 0, y = 0 are handy).

Step 2

Plot the points on the same axes and draw both lines.

Step 3

Read the common point: its coordinates are the solution.

Step 4

If the lines are parallel there is no solution; if they coincide there are infinitely many.

Pair	Graph	Result
x + 3y = 6, 2x − 3y = 12	Meet at (6, 0)	Unique: x = 6, y = 0
x + 2y − 4 = 0, 2x + 4y − 12 = 0	Parallel (ratios ½ = ½ ≠ ⅓)	No solution
2x + 3y = 9, 4x + 6y = 18	Same line	Infinitely many

Using the Ideas

- **Celsius–Fahrenheit:** F = (9/5)C + 32. 30 °C = 86 °F, 95 °F = 35 °C, and the two scales agree at **−40**.
- **Mobile plans:** A = ₹50 + 0.20 per minute, B = ₹30 + 0.30 per minute. Equal when 50 + 0.2t = 30 + 0.3t, i.e. at t = 200 minutes; A is cheaper beyond that.
- **x-intercept** of y = mx + d is **−d/m** (m ≠ 0).

Likely question	Model answer
What is the slope of a line through (x_1, y_1) and (x_2, y_2)?	$(y_2 - y_1)/(x_2 - x_1)$, and it is the same for every pair of points on the line.
How can you tell without graphing that a pair has no solution?	$a_1/a_2 = b_1/b_2 \neq c_1/c_2$: the lines are parallel.

Likely question	Model answer
Why can a linear equation in two variables not have only one solution?	For every value of x we get a matching y, so there are infinitely many solutions forming a line.
What does c = 0 tell you about the line?	It passes through the origin.
Which method suits 9x + 7y = 107 and 7x + 9y = 101?	Add and subtract the equations to get x + y and x − y.

★ KEY TAKEAWAYS

- A linear equation **ax + by + c = 0** has **infinitely many** solutions, whose graph is a **straight line**.
- **Slope = rise/run = $(y_2 - y_1)/(x_2 - x_1)$**; in **y = mx + d**, m is the slope and d the y-intercept.
- Horizontal lines have slope **0**; vertical lines have **undefined** slope.
- A pair of linear equations is solved by **substitution**, **elimination** or **graphs**, and the answer is checked in both equations.
- **$a_1/a_2 \neq b_1/b_2$** → unique solution; **$= c_1/c_2$** → infinitely many; **$\neq c_1/c_2$** → no solution.
- In word problems define variables, form **two equations**, solve and **verify with the story**.

CHAPTER 14 · MATHS (PART 2)

Math of Space: Surface Area and Volume

Solids have a surface that can be covered and an inside that can be filled. This chapter gives the formulas for the surface area and volume of the standard solids, shows where they come from, and uses them to solve everyday problems and to make sensible estimates.

Big Questions

- How do we measure the surface and the space occupied by a solid?
- How are the formulas for cuboid, cylinder, cone, pyramid and sphere related?
- How can we estimate quantities when exact data are not available?

Cuboids and Cubes

DEFINITION

A **cuboid** has six rectangular faces and is described by its length **l**, width **w** and height **h**. A **cube** is a cuboid with l = w = h = **a** (the side). **Volume** is the number of unit cubes that fit inside; it is measured in cubic units (cm³, m³).

Solid	Total surface area (TSA)	Volume
Cuboid	2(lw + wh + lh)	lwh
Cube	$6a^2$	a^3

- Think of a cuboid as many thin rectangular sheets stacked like a pack of cards: **volume = area of base × height**.
- The cube formulas are special cases of the cuboid formulas with l = w = h.
- **Unit conversion:** 1 m³ = 1000 litres, 1000 cm³ = 1 litre.

▸ Same volume, different surface area

A cube of side 6 cm and a cuboid 9 cm × 6 cm × 4 cm both have volume **216 cm³**, but their surface areas are **216 cm²** and **228 cm²**. For the same capacity a cube needs less covering material, while a cuboid exposes more area to its surroundings. This idea is used in packaging, cooling and engineering.

IMPORTANT

If a cube is cut into many small cubes, the volume stays the same but the total surface area increases greatly. Cutting a 4 cm cube into 1 cm cubes increases the surface area from 96 cm² to 384 cm².

▸ Painted-cube counting (side n, cut into unit cubes, n ≥ 2)

Painted faces	Number of small cubes
3 (corners)	8
2 (edges)	12(n − 2)
1 (faces)	$6(n-2)^2$

Painted faces	Number of small cubes
0 (inside)	$(n-2)^3$

Right Circular Cylinder

DEFINITION

In a **right circular cylinder** the cross-section parallel to the base is a circle and the axis is **perpendicular** to the base. A **cross-section** is the shape obtained on taking a very thin slice of a solid; a slice parallel to the axis of a cylinder is a rectangle.

- Cut the curved surface along a line parallel to the axis and open it out: it is a **rectangle** of sides $2\pi r$ and h.

Quantity	Formula
Curved surface area (CSA)	$2\pi rh$
TSA, closed at both ends	$2\pi rh + 2\pi r^2 = 2\pi r(h + r)$
Surface area, open at one end (tumbler)	$2\pi rh + \pi r^2 = \pi r(2h + r)$
Volume	$\pi r^2 h$ (base area × height)

Imagine a cylinder as a stack of thin circular discs (like khakhras); the volume is the base area times the height.

▸ Worked example

Step 1
Read $r = 3.5$ cm and $h = 25$ cm for a cylindrical kaleidoscope; use $\pi = 22/7$.

Step 2
Only the curved surface needs chart paper, so CSA $= 2\pi rh$.

Step 3
CSA $= 2 \times 22/7 \times 3.5 \times 25 =$ **550 cm²**.

NOTE

Ratios: if r and h change, CSA depends on **r × h** and volume depends on **r² × h**. If r is doubled and h is halved, CSA stays the same while the volume doubles.

Cones

DEFINITION

A **right circular cone** is formed when a right triangle is rotated about one of its perpendicular sides. The radius is **r**, the height is **h** and the slant height is **l**, where **$l^2 = h^2 + r^2$**.

- Cut a cone along a slant edge and unroll it: you get a **sector of a circle** of radius l whose arc length equals the base circumference $2\pi r$.
- Cutting the sector into thin triangles of height l gives CSA = ½ × circumference × l.

Quantity	Formula
Curved surface area	πrl
Total surface area	$\pi rl + \pi r^2 = \pi r(l + r)$
Volume	$\frac{1}{3}\pi r^2 h$

REMEMBER

A cone has **one-third** the volume of a cylinder with the same base and height. Experiment: fill the cone with salt and empty it into the cylinder; it takes exactly **three** fills. The factor ⅓ is proved with calculus in Grade 12.

▸ Worked example — triangle rotated about a leg

Step 1
Triangle 5, 12, 13: $5^2 + 12^2 = 13^2$, so it is right-angled.

Step 2
Rotate about the side 12 cm: height h = 12, radius r = 5.

Step 3
Volume = $\frac{1}{3}\pi \times 25 \times 12 =$ **100π cm³**.

Pyramids

A **pyramid** joins every point of a flat base (triangle, square or any shape) to a single point, the **apex**, that lies in a different plane. Examples: the pyramids of Egypt and mountain peaks such as Shivling Peak in Uttarakhand.

Quantity	Rule
Volume	⅓ × area of base × height (any base shape)
Surface area	Add the areas of all the faces — no separate formula is needed

NOTE

A cone is a pyramid with a circular base; both use the same ⅓ factor.

Spheres and Hemispheres

DEFINITION

A **sphere** is the set of all points in space at a fixed distance r (the radius) from a fixed point O (the centre). A **hemisphere** is half a sphere, cut by a plane through the centre; it has one curved surface and one circular base.

Solid	Curved / total surface area	Volume
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	CSA = $2\pi r^2$; TSA = $3\pi r^2$ (curved + base πr^2)	$\frac{2}{3}\pi r^3$

- **Archimedes' cylinder:** a cylinder of radius r and height 2r tightly fits over a sphere; its curved surface area $2\pi r \times 2r = 4\pi r^2$ equals the sphere's surface area. (This idea is used in rectangular world maps.)

- **Link with cones:** divide the surface into tiny regions and join them to the centre to get many thin cones of height r, so $V = \frac{1}{3} \times (\text{surface area}) \times r = \frac{1}{3} \times 4\pi r^2 \times r = \frac{4}{3}\pi r^3$.
- **String experiment:** string wound over a ball fills four circles of the same radius, verifying $4\pi r^2$.

Formula Table

3D shape	Surface area	Volume
Cuboid	$2(lw + wh + lh)$	lwh
Cube	$6a^2$	a^3
Cylinder	CSA $2\pi rh$; TSA $2\pi r(h + r)$	$\pi r^2 h$
Cone	CSA πrl; TSA $\pi r(l + r)$	$\frac{1}{3}\pi r^2 h$
Pyramid	Sum of face areas	$\frac{1}{3} \times$ base area $\times h$
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	CSA $2\pi r^2$; TSA $3\pi r^2$	$\frac{2}{3}\pi r^3$

IMPORTANT

Use **$\pi = 22/7$** (or 3.14) as the question directs. Always write units: cm^2, m^2 for areas and cm^3, m^3 for volumes.

Scaling and Melting-Recasting

- If every length is multiplied by **k**, surface area is multiplied by **k^2** and volume by **k^3** (for example, doubling a sphere's radius gives 4 times the area and 8 times the volume).
- When a solid is **melted and recast**, the volume stays the same: number of pieces = volume of the original ÷ volume of one piece.
- When a solid sinks in water in a container, the **displaced volume** = volume of the solid = base area of container × rise in level.

Areas and Volumes Around Us: Guesstimates

A **guesstimate** is a reasoned estimate made when exact data are not available, using assumptions, approximations and everyday knowledge. The answer depends on the assumptions, so different sensible assumptions give different answers.

Step 1
Guess first (hundreds? thousands?).

Step 2
List the assumptions, for example the radius of a golgappa and the fraction of it filled with pani.

Step 3
Compute the volumes with the formulas.

Step 4
Divide the total volume by the volume of one item and round sensibly.

Step 5
State the assumptions with the answer.

Example from the chapter	Idea
Golgappas from a vessel of pani	Cylinder volume ÷ pani in one sphere
Tennis balls in a classroom	Room volume ÷ ball volume (balls do not fill all the space)
Water on the Earth as a thin layer	Volume = surface area × thickness

Typical Exam Questions

Likely question	Model answer
Find the TSA of a cube of volume 64 cm³.	Side = 4 cm, so TSA = 6 × 16 = 96 cm².
A cone has r = 7 cm and h = 24 cm. Find its slant height and CSA (π = 22/7).	l = √(49 + 576) = 25 cm; CSA = πrl = 22/7 × 7 × 25 = 550 cm².
Find the volume of a hemisphere of radius 10.5 cm (π = 22/7).	⅔ × 22/7 × 10.5³ = 2425.5 cm³.
Why is the volume of a pyramid ⅓ × base area × height?	A pyramid has one-third the volume of a prism with the same base and height, just as a cone has one-third the volume of a cylinder.
A metal sphere of radius 3 cm is melted into cones of radius 1 cm and height 3 cm. How many cones?	Sphere = 36π cm³; one cone = π cm³; so 36 cones.

★ KEY TAKEAWAYS

- Cuboid: **TSA = 2(lw + wh + lh)**, **V = lwh**; cube: **6a²** and **a³**.
- Cylinder: **CSA = 2πrh**, **TSA = 2πr(h + r)**, **V = πr²h**.
- Cone: **l² = h² + r²**, **CSA = πrl**, **V = ⅓πr²h** (one-third of the cylinder).
- Pyramid: **V = ⅓ × base area × height**; surface area is the sum of the faces.
- Sphere: **4πr²** and **⁴⁄₃πr³**; hemisphere: **CSA 2πr², TSA 3πr², V ⅔πr³**.
- Scaling by k multiplies area by k² and volume by k³; melting and recasting keeps volume the same.
- Guesstimates need clear assumptions and a sensible rounded answer.